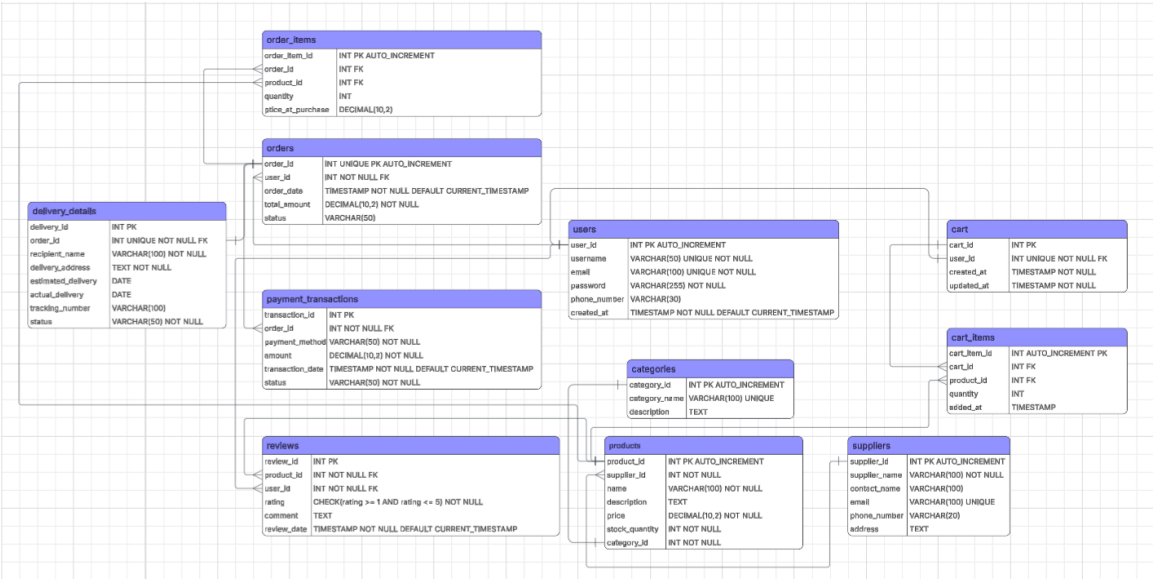
Online Shopping Platform

1.1 Logical Scheme



1.2 Objects

The following section describes each table in the database including its fields, purpose, and constraints.

Table: users

Field Name	Field Description	Data Type / Constraints
user_id	Unique identifier of the user	INT PK AUTO_INCREMENT
username	User login name	VARCHAR(50) NOT NULL
email	User email address	VARCHAR(100) UNIQUE NOT NULL
password	User password hash	VARCHAR(255) NOT NULL
phone_number	User phone number	VARCHAR(30)
created_at	Account creation timestamp	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Table: suppliers

Field Name	Field Description	Data Type / Constraints
supplier_id	Unique supplier identifier	INT PK AUTO_INCREMENT
supplier_name	Supplier company name	VARCHAR(100) NOT NULL
contact_name	Supplier contact person	VARCHAR(100)
email	Supplier email address	VARCHAR(100) UNIQUE
phone_number	Supplier phone number	VARCHAR(20)
address	Supplier address	TEXT

Table: categories

Field Name	Field Description	Data Type / Constraints
category_id	Unique category identifier	INT PK AUTO_INCREMENT
category_name	Category name	VARCHAR(100) UNIQUE
description	Description of category	TEXT

Table: products

Field Name	Field Description	Data Type / Constraints
product_id	Unique product identifier	INT PK AUTO_INCREMENT
supplier_id	Supplier providing the product	INT FK
name	Product name	VARCHAR(100) NOT NULL
description	Product description	TEXT
price	Product price	DECIMAL(10,2) NOT NULL
stock_quantity	Number of items available	INT NOT NULL
category_id	Category of product	INT FK

Table: cart

Field Name	Field Description	Data Type / Constraints
cart_id	Unique cart identifier	INT PK
user_id	Owner of the cart	INT FK UNIQUE
created_at	Date cart was created	TIMESTAMP
updated_at	Last cart update	TIMESTAMP

Table: cart_items

Field Name	Field Description	Data Type / Constraints
cartItem_id	Unique cart item identifier	INT PK AUTO_INCREMENT
cart_id	Cart reference	INT FK
product_id	Product in cart	INT FK
quantity	Number of products in cart	INT
added_at	Date item added to cart	TIMESTAMP

Table: orders

Field Name	Field Description	Data Type / Constraints
order_id	Unique order identifier	INT PK AUTO_INCREMENT
user_id	User who placed order	INT FK
order_date	Date when order was created	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
total_amount	Total cost of the order	DECIMAL(10,2) NOT NULL
status	Current order status	VARCHAR(50)

Table: order_items

Field Name	Field Description	Data Type / Constraints
order_item_id	Unique order item identifier	INT PK AUTO_INCREMENT
order_id	Order reference	INT FK
product_id	Product included in order	INT FK
quantity	Quantity of the product	INT
price_at_purchase	Product price at purchase time	DECIMAL(10,2)

Table: payment_transactions

Field Name	Field Description	Data Type / Constraints
transaction_id	Unique payment transaction identifier	INT PK
order_id	Order being paid	INT FK
payment_method	Method of payment	VARCHAR(50)
amount	Payment amount	DECIMAL(10,2)
transaction_date	Date of payment	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
status	Transaction status	VARCHAR(50)

Table: delivery_details

Field Name	Field Description	Data Type / Constraints
delivery_id	Unique delivery identifier	INT PK
order_id	Related order	INT FK
recipient_name	Person receiving the order	VARCHAR(100)
delivery_address	Delivery address	TEXT
estimated_delivery	Estimated delivery date	DATE

actual_delivery	Actual delivery date	DATE
tracking_number	Shipment tracking code	VARCHAR(100)
status	Delivery status	VARCHAR(50)

Table: reviews

Field Name	Field Description	Data Type / Constraints
review_id	Unique review identifier	INT PK
product_id	Reviewed product	INT FK
user_id	User who wrote review	INT FK
rating	Rating score (1–5)	CHECK(rating >= 1 AND rating <= 5)
comment	Review text	TEXT
review_date	Date review created	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Comments on Table Relationships

users -> orders : one user can create many orders
orders -> order_items : one order contains multiple products
products -> categories : each product belongs to one category
products -> suppliers : each product has one supplier
users -> cart : each user has one cart
cart -> cart_items : cart contains multiple products
orders -> payment_transactions : orders can have payment records
orders -> delivery_details : each order has delivery information
users + products -> reviews : users can review products

Example Data

order_id	username	total_amount	delivery_address	status
1	Kairat_Roman	1200	67 Nursaya	Paid
2	Lisa_Lisa	599.9	20 Avangard	Paid